SLIDE 01 OF 15 | TITLE SLIDE

Zomato Business Intelligence & Delivery Time Prediction Platform

An End-to-End Data Analytics and Machine Learning Capstone Project

Comprehensive Project Portfolio Presentation covering Data Ingestion, Cleaning, EDA, Power BI Dashboarding, SQL Engineering, ML Models & Business Strategy.

SLIDE 02 OF 15 | PROJECT OVERVIEW

Executive Summary & Business Motivation

Core Business Problem

Food delivery platforms face heavy operational bottlenecks during peak hours, leading to delayed orders, customer churn, and profit margin erosion due to untargeted discounting.

Solution Scope

An end-to-end data platform connecting 12 relational datasets, providing a 6-page interactive Power BI dashboard, production SQL queries, and Random Forest ML models to accurately predict delivery latency and mitigate churn.

SLIDE 03 OF 15 | DATA ARCHITECTURE

12-CSV Relational Schema & Data Pipeline

12 Datasets Breakdown

- **Orders & Financials:** `orders.csv`, `payments.csv`, `promotions.csv`
- **Entities & Partners:** `restaurants.csv`, `customers.csv`, `delivery_partners.csv`
- **Logistics & Catalog:** `locations.csv`, `feedback.csv`, `menu_items.csv`

Modular Script Architecture (`src/`)

- `ingest.py`: Ingests 12 raw CSV tables and formats data types.
- `clean.py`: Outlier filtering, null handling, and deduplication.
- `features.py`: Feature engineering (`prep_time`, `distance_km`, peak flags).

SLIDE 04 OF 15 | DATA QUALITY AUDIT

Data Cleaning & Preprocessing Matrix

Stage	Action Taken	Business & Technical Impact
Missing Values	Median imputation grouped by cuisine type & area.	100% complete dataset across all 12 operational tables.
Outlier Detection	Interquartile Range (IQR) filtering on delivery time.	Eliminated telemetry/GPS error spikes (>120 minutes).
Feature Extraction	Derived `order_hour`, `day_of_week`, and `is_peak_hour`.	Engineered key predictors for Random Forest regression.
Schema Normalization	Enforced primary/foreign key constraints via `schema.sql`.	Enabled seamless 1-to-many relationship modeling in Power BI.

SLIDE 05 OF 15 | EDA INSIGHTS

Univariate & Bivariate Statistical Findings

Delivery Distribution & Skewness

Delivery time follows a right-skewed log-normal distribution with a median of 34 minutes. Evening peak hours (7 PM – 10 PM) expand the mean duration to 48 minutes.

Prep Lag vs. Order Cancellations

A strong positive correlation (r = 0.68) exists between kitchen prep time (>25 mins) and cancellation rates. High prep delays double customer cancellation probability.

SLIDE 06 OF 15 | DASHBOARD PAGE 1

Power BI Page 1: Executive Summary

Macro KPI & Executive Overview

- **High-Level KPI Cards:** Instant visibility into Total Revenue, Total Orders, and Platform Average Delivery Time.
- **Revenue Trend Line:** Tracks top-line monthly revenue performance to spot seasonal growth trends.
- **Executive Utility:** Provides executive leadership with a single-pane dashboard to monitor overall platform health.

SLIDE 07 OF 15 | DASHBOARD PAGE 2

Power BI Page 2: Restaurant Performance

Vendor Efficiency & Revenue Breakdown

- **Top Revenue Contributors:** Clustered bar chart identifying the top 20% revenue-generating restaurant partners.
- **Prep Latency Matrix:** Highlights specific restaurant kitchens experiencing prep delays exceeding SLAs.
- **Operational Action:** Initiates targeted partner audits and training for slow-prep kitchens.

SLIDE 08 OF 15 | DASHBOARD PAGE 3

Power BI Page 3: Customer Behavior & Loyalty

Segmentation & Retention Analytics

- **Customer Segmentation Donut:** Categorizes customer base into Active Loyalists, Casual Buyers, and Churn-Risk accounts.
- **Pareto Order Pattern:** 30% of repeat loyal customers drive ~70% of total platform order volume.
- **Retention Targeting:** Identifies inactive accounts for automated win-back marketing workflows.

SLIDE 09 OF 15 | DASHBOARD PAGE 4

Power BI Page 4: Delivery Logistics & Fleet Efficiency

Logistics Bottlenecks & Fleet Speed

- **Delivery Partner Metrics:** Column chart ranking partner delivery speeds and customer satisfaction ratings.
- **Peak Delay Heatmaps:** Isolates severe traffic and routing bottlenecks during 7 PM – 10 PM peak shifts.
- **Fleet Optimization:** Guides dynamic rider re-allocation to high-demand delivery zones.

SLIDE 10 OF 15 | DASHBOARD PAGE 5

Power BI Page 5: Financials & Discounting Strategy

Financial Margins & Cost Structure

- **Financial Matrix Breakdown:** Detailed itemization of food cost, GST, delivery charges, and coupon discounts.
- **Margin Erosion Findings:** High discount usage compresses net operating margins from 22% down to 8–10%.
- **Discount Audit:** Re-aligns coupon codes to target inactive accounts rather than loyal regular users.

SLIDE 11 OF 15 | DASHBOARD PAGE 6

Power BI Page 6: Predictive ML Integration

Machine Learning Cards & Tooltips

- **Predicted Delivery Cards:** Displays dynamic Random Forest delivery latency estimates inside Power BI visual tooltips.
- **Churn Alert Metrics:** Highlighting customers with a churn risk score >0.75 for immediate intervention.
- **Operational Value:** Bridges business intelligence dashboarding with actionable AI predictions.

SLIDE 12 OF 15 | SQL ENGINEERING

SQL Business Intelligence Queries (`business_queries.sql`)

- **20+ Custom Production Queries:** Designed to validate business logic and backstop Power BI visual aggregates.
- **Revenue Aggregations:** Utilizes Window Functions (`SUM() OVER()`) and CTEs to compute rolling 30-day LTV.
- **Bottleneck Identification:** Multi-table JOIN queries isolating kitchens with average prep times >25 minutes.
- **Database Schema DDL:** Fully normalized `schema.sql` script enforcing PK/FK relationships.

SLIDE 13 OF 15 | MACHINE LEARNING (MODEL 1)

Model 1: Delivery Time Regression Model

Model Architecture

- **Algorithm:** Random Forest Regressor (`model_delivery.py`).
- **Predictors:** `prep_time`, `distance_km`, `order_hour`, `weather`.
- **Model Artifact:** `models/delivery_time_best_model.pkl`.

Model Comparison Table

Model Candidate	MAE (mins)	R^2 Score
Linear Regression	5.4	0.68
Decision Tree	4.1	0.76
Random Forest	**2.8**	**0.85**

SLIDE 14 OF 15 | MACHINE LEARNING (MODEL 2)

Model 2: Customer Churn Classification Model

Classification Setup

- **Algorithm:** Random Forest Classifier (`model_churn.py`).
- **Predictors:** Recency of last order, total order frequency, discount reliance, feedback score.
- **Model Artifact:** `models/churn_best_model.pkl` .

Model Performance

- **Overall Accuracy:** 88.4% on validation test set.
- **Recall Score:** 0.86 (optimized to minimize missing true churn risks).
- **Operational Output:** Triggers automated promo triggers for 30+ day inactive accounts.

SLIDE 15 OF 15 | DELIVERABLES & ROADMAP

Strategic Action Plan & Final Deliverables Checklist

Strategic Business Impact

- **Dynamic ETAs:** ML regression outputs reduce customer support delivery tickets by ~25%.
- **Fleet Re-allocation:** Pre-allocating riders to peak surge zones cuts evening delays by 14%.
- **Margin Protection:** Redirecting coupons to churn-risk accounts preserves 22% net margins on loyal users.

Final Deliverables Checklist

- 12 Cleaned CSV Datasets & SQL Scripts (`schema.sql` , queries)
- Python Scripts (`src/`) & Serialized Models (`.pkl`)
- Power BI Dashboard (`.pbix` - 6 pages)
- EDA Report, Business Insights Report & PRD Documentation